

LTC3892-1 Supply Design Summary Report

Vin : 25V (min.), 31V (nom.), 37V (max.)

Output Rails : Vout1 = 23.789V / 37A (max.)

Project Name :

Project Date :

Designer :

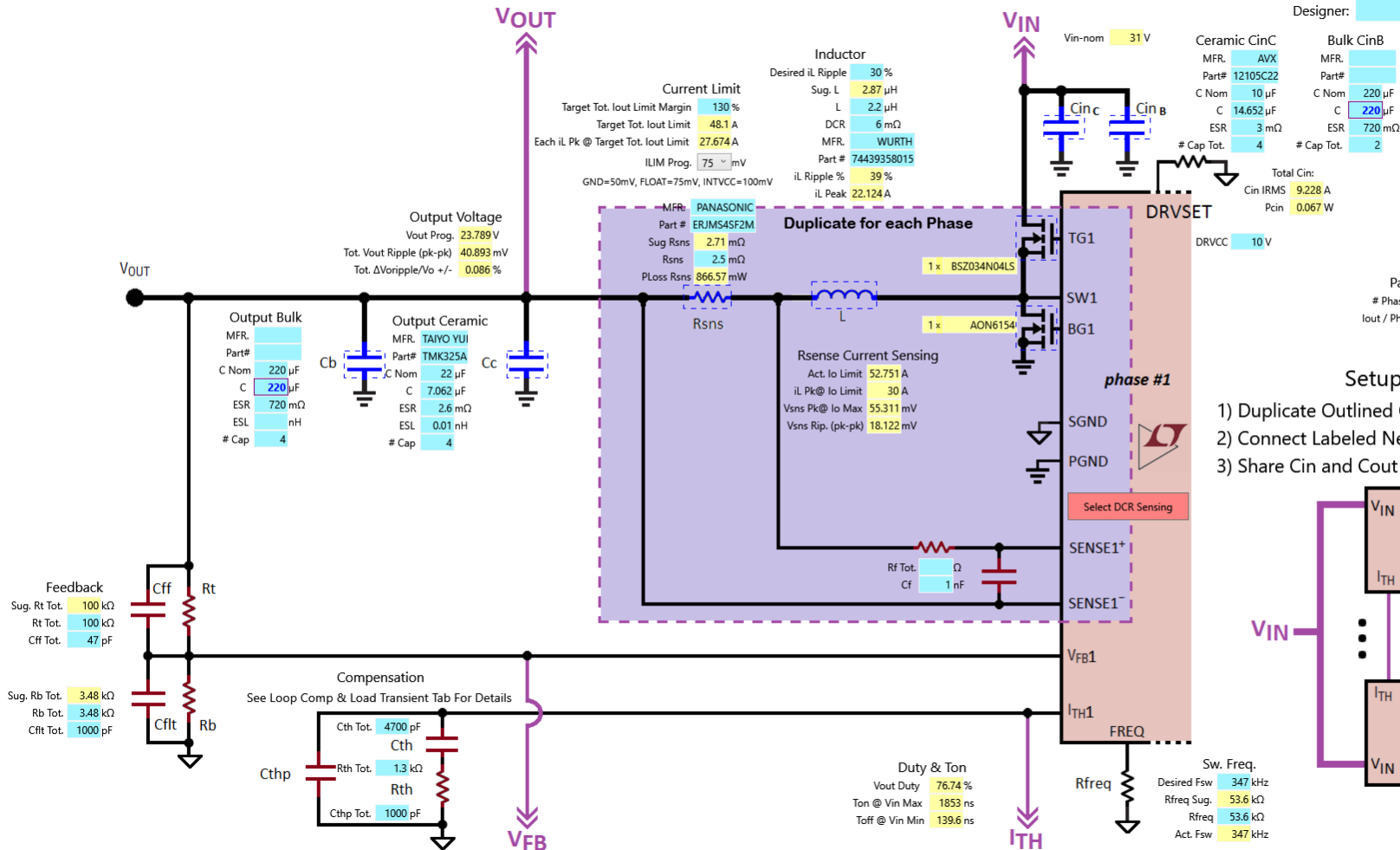


LTC3892-1 Solution - Simplified Schematic

Vin : 25V (min.), 31V (nom.), 37V (max.)

Output Rails : Vout1 = 23.789V / 37A (max.)

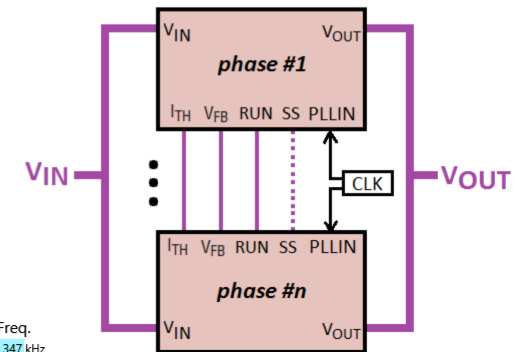
LTC3892-1 - 60Vin Low Iq, 2-Phase Synchronous Step-Down DC/DC Controller



Project Name:
Date:
Designer:

Setup For Paralleling :

- 1) Duplicate Outlined Components For Each Phase
- 2) Connect Labeled Nets Among Paralleled Phases
- 3) Share Cin and Cout

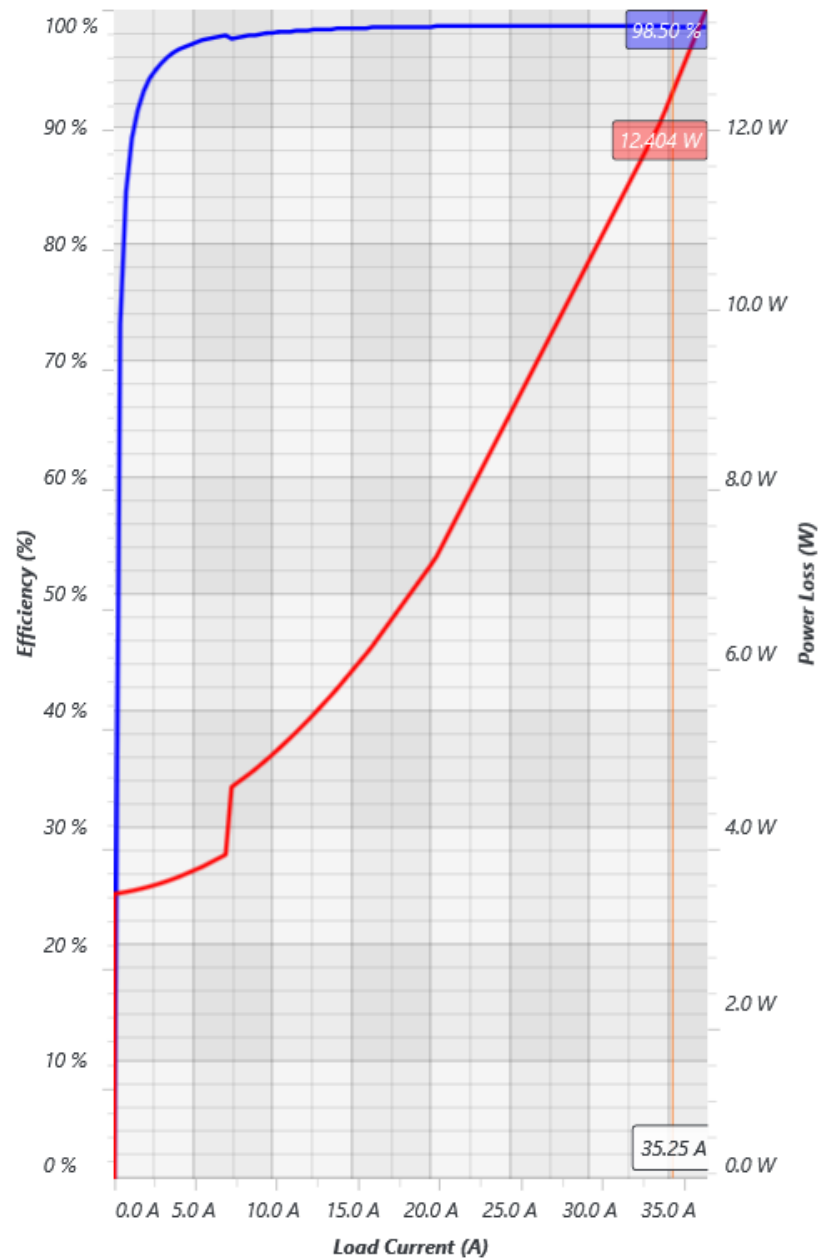


LTC3892-1 Solution - Efficiency & Loss Estimations

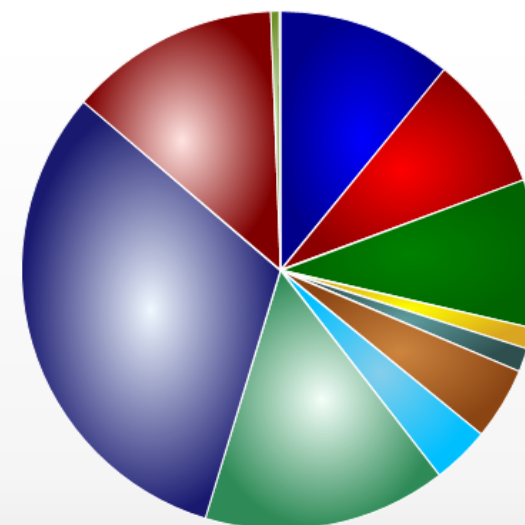
Rail # 1 : $V_{in} = 31V$, $V_{out1} = 23.789V$

* Estimations For CCM Mode Only. Inductor AC Losses Entered by User

Rail #1 (23.789V) Efficiency & Power Loss



Rail #1 (23.789V) Power Loss Breakdown (Full Load)



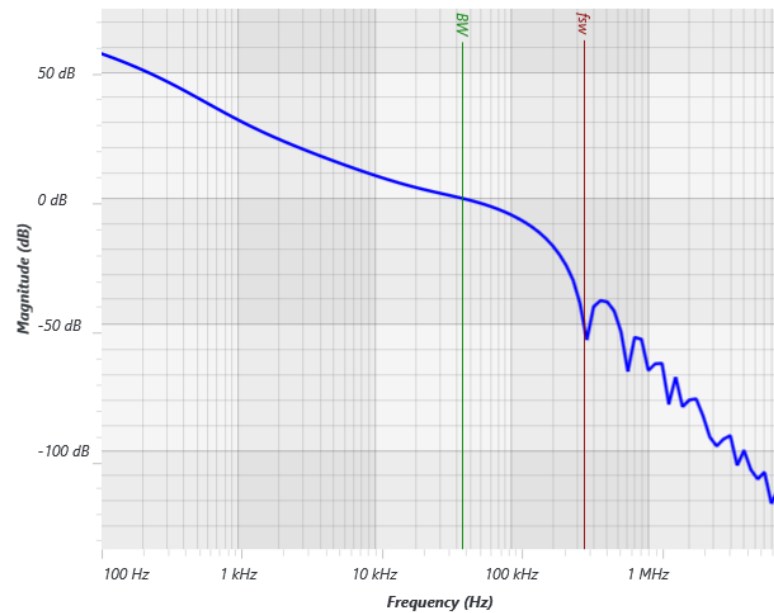
- Control Fet Conduction (1.436W,10.89%)
- Control Fet Turn On (1.115W,8.46%)
- Control Fet Turn Off (1.215W,9.22%)
- Control Fet Driving (0.174W,1.32%)
- Sync Fet Conduction (0.194W,1.47%)
- Sync Fet Driving (0.597W,4.53%)
- Sync Fet Body Diode (0.476W,3.61%)
- IC LDO (2.003W,15.19%)
- Inductor DCR (4.16W,31.56%)
- Rsense (1.733W,13.15%)
- Cin Losses (0.07W,0.53%)
- Cout (0.01W,0.08%)

LTC3892-1 Solution - Loop Gain & Load Transient Estimations

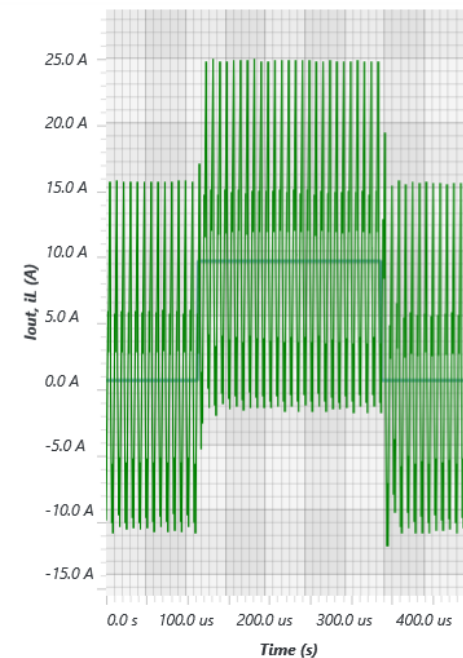
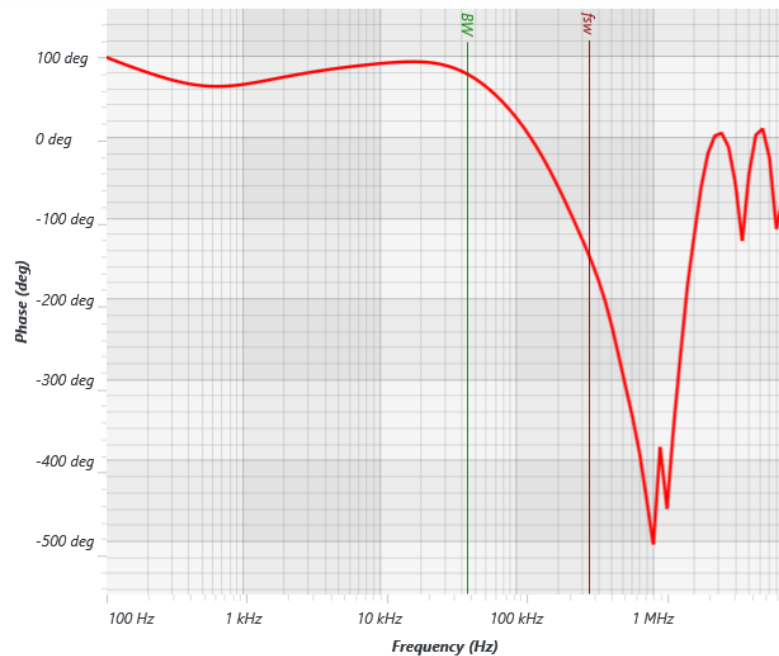
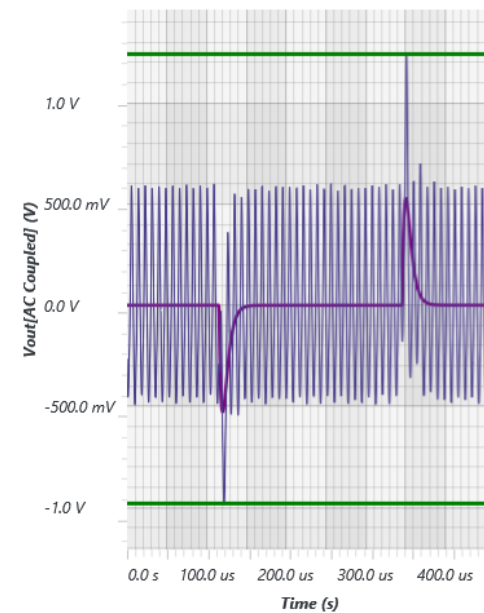
Rail # 1 : $V_{in} = 31V$, $V_{out1} = 23.789V$, $I_{out1} = 37A$

** Estimations For CCM Mode Only. Estimations Based On Small Signal Avg. Model*

Rail #1 (23.789V) Loop Gain



Rail #1 (23.789V) Load Transient



LTC3892-1 Solution - Summary

LTC3892-1 Supply Design Summary



Project Info:

Design Specifications

Steady State :

Rail #	Vin Min.	Vin Nom.	Vin Max.	Fsw	Vo	ΔVo	ΔVo%	Io Max	ΔILp-p	ΔIL%	iLpk	Duty	Ton min.	Toff min.
1	25 V	31 V	37 V	347 kHz	23.79 V	40.89 mV	0.09 %	37 A	7.25 A	39 %	22.12 A	76.74 %	1853 ns	140 ns

Efficiency and Loop :

Rail #	Vo	Iomax	Eff.@Iomax	PLoss@Iomax	Loop BW	Loop PM	Step Low	Step High	Step Slew	ΔVo@Step	ΔVo@Step %
1	23.79 V	37 A	98.52 %	13.18 W	44.67 kHz	77.52 deg	0 A	9.25 A	9.999999 A/μs	1236 mV	+/-5.2 %

Recommendations and Warnings :

Message
Rail #1 Capacitance value is not de-rated. Inadequate information provided to estimate de-rated value. Please check the capacitor's entries in library.

Power Components

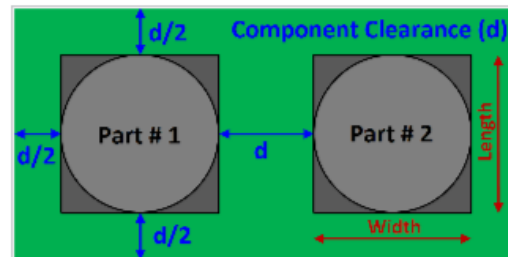
Power Components Bill Of Materials :

Export BOM

Ref. Des.	Value	Quantity	Description	Mfr. Name	Mfr. Part #	Pkg. (Imperial)	L(mm)	W(mm)	H(mm)	User Note
U1		1	IC	LINEAR TECH	LTC3892-1		5	5	0.8	
Lo1 Lo2	2.2μH	2	IND	WURTH	74439358015		9	8.5	8	
Cinb1 Cinb2	220μF	2	CAP				0	0	0	
Cinc1 Cinc2 Cinc3 Cinc4	10μF	4	CAP	AVX	12105C226KAT2A	1210	3.2	2.5	1.7	
Cob1 Cob2 Cob3 Cob4	220μF	4	CAP				0	0	0	
Coc1 Coc2 Coc3 Coc4	22μF	4	CAP	TAIYO YUDEN	TMK325ABJ476MM-P	1210	3.2	2.5	2.8	
Rsense1 Rsense2	2.5mΩ	2	RES	PANASONIC	ERJMS45F2M2U		6.4	3.2	0.64	
Mtop1 Mtop2	40V	2	FET	INFINEON	BSZ034N04LS	PG-TSDSON-8	3.4	3.4	1.1	
Mbot1 Mbot2	45V	2	FET	AOS	AON6154		6.05	5.2	0.95	

Power Components Footprint :

# Components	23	
Max. Height	8	mm
Component Clearance (d)	1.5	mm
* Power Components Area (Excludes ICs)	583.8	mm^2
	0.905	in^2
* Power Components Area (Includes ICs)	626.1	mm^2
	0.97	in^2



* Notes :

1. The calculated power component area is only the simple sum of component footprint areas with given clearance, assuming all power components are on the same side of PCB. It is NOT the final PCB size with layout design.
2. Component count should change with the number of paralleled phases.